



Recommender Systems



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Agenda

Background

Getting started

Part 1 (~30 min)

- Exploratory data analysis
- Popularity-based recommendations

Part 2 (~40 min)

- Tensorflow Extended Pipelines
- Two tower architecture

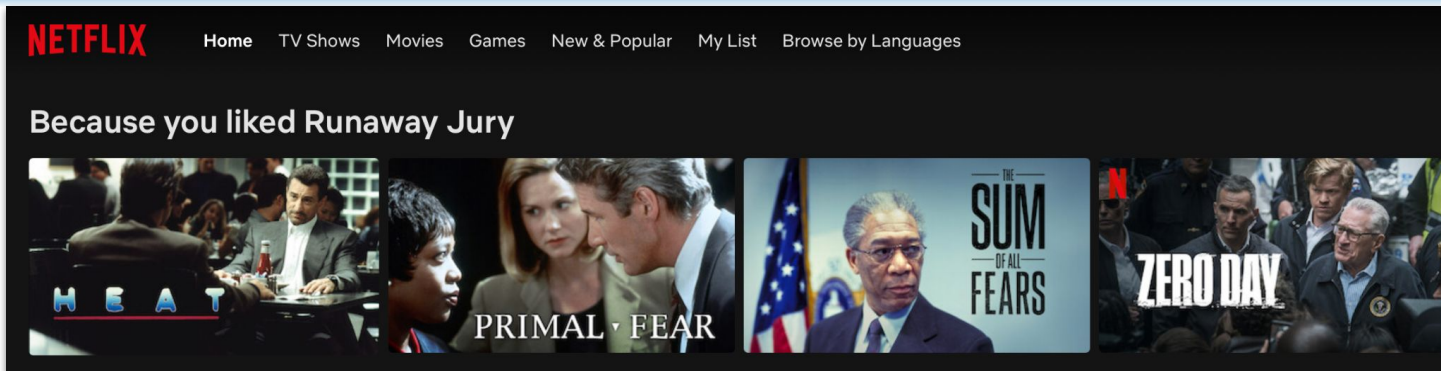


Part 3 (~60 min)

- Experimentation



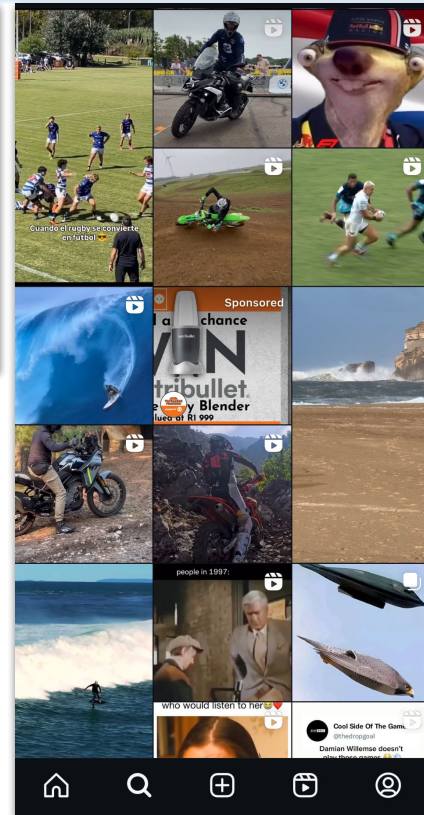
What are Recommender Systems?



amazon



You Tube



Recommendations in eCommerce

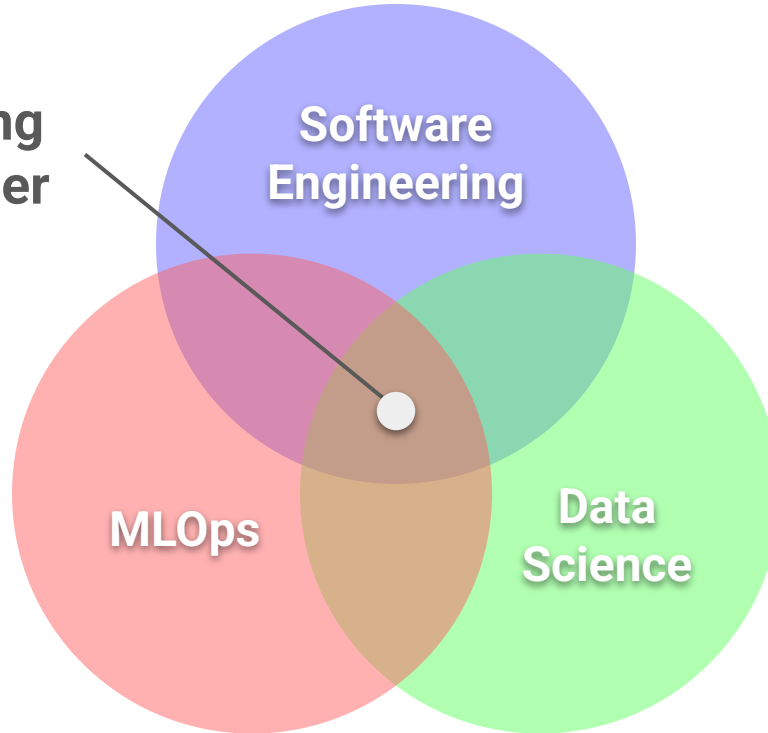
Recommendations drive

35%

of **amazon** sales

Architecture Considerations

**Machine Learning
Software Engineer**



Getting Started

Find the repo, and follow the **Getting Started** instructions in the README.

- <https://github.com/stefandominicus-takealot/RecSys2025>

Apple Silicon! If you have a newer Mac, we'll need to spin up a VM for you.

Once you're done, please help someone else 🧐

Part 1:

Top Rated Products



LEGO® Speed Champions Oracle Red Bull Racing RB20 F1® Race Car Vehicle Set 77243

LEGO

Build your own Oracle Red Bull Racing RB20 F1® Race Car with this LEGO® Speed Champions building set for adults and Formula 1® fans.

★ 4.9 30 Reviews

In stock CPT DBN JHB [Learn More](#)

R 458

Estimated Delivery CPT DBN JHB

Get It Tomorrow! [T&Cs Apply](#)

FREE DELIVERY WITH [TAKEALOTMORE](#)

+ Add to Cart

♥ Add to List

Reviews

4.9 ★★★★★ 30 Reviews



Write Review

Filter by Ratings

- ☐ 5 ★
- ☐ 4 ★
- ☐ 3 ★
- ☐ 2 ★
- ☐ 1 ★

1 to 10 of 30 Reviews

Sort by: Most Helpful

★★★★★

Laricha - 13 Mar 2025 (Reviewed 7 days after purchase)

Authentic Design: Captures the sleek and aggressive look of the RB20 with incredible attention to detail. Working Mechanics: Functional steering, suspension, and engine movement make it more than just a display piece. High-Quality Stickers & Branding: Official Red Bull and F1 team details add to the realism. Sturdy & Well-Balanced: Feels solid once built, making it great for both display and light play.

[Read More](#)

👍 Helpful (2)

[Report Review](#)

★★★★★

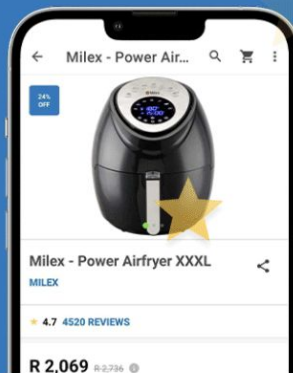
Nathaniel Chad - 17 Mar 2025 (Reviewed 7 days after purchase)

Love the product! Fast next day delivery!

👍 Helpful (1)

[Report Review](#)

rate & review your products



Hi Stefan,

Thank you for shopping on takealot.com!

We'd love to know your honest thoughts on your recently purchased items.



Plus to reward loyal shoppers, we're giving away a **R1000 Takealot voucher** every month to one lucky reviewer!

[T&Cs apply.](#)

Ratings Aggregation Notebook

```
RecSys2025
```

```
> recommender_systems
```

```
> notebooks
```

```
> ratings_aggregation.ipynb
```



Part 2:

Robust Build Pipelines

Tensorflow Extended Pipelines

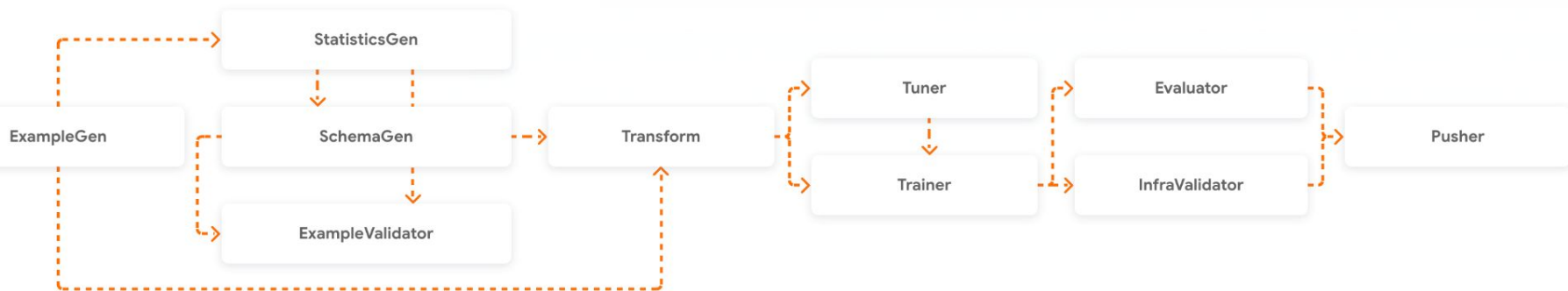
How it works

A TFX pipeline is a sequence of components that implement an ML pipeline which is specifically designed for scalable, high-performance machine learning tasks. Components are built using TFX libraries which can also be used individually.

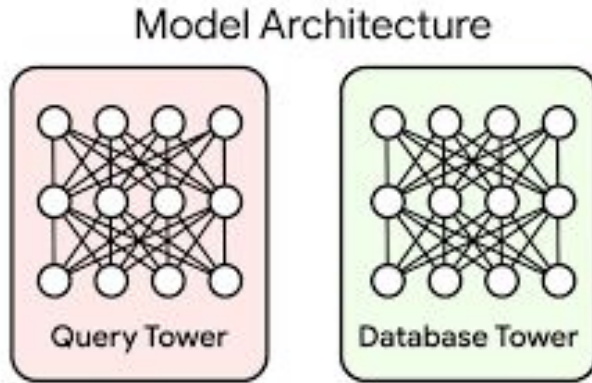
Ingest & validate data

Train & analyze model

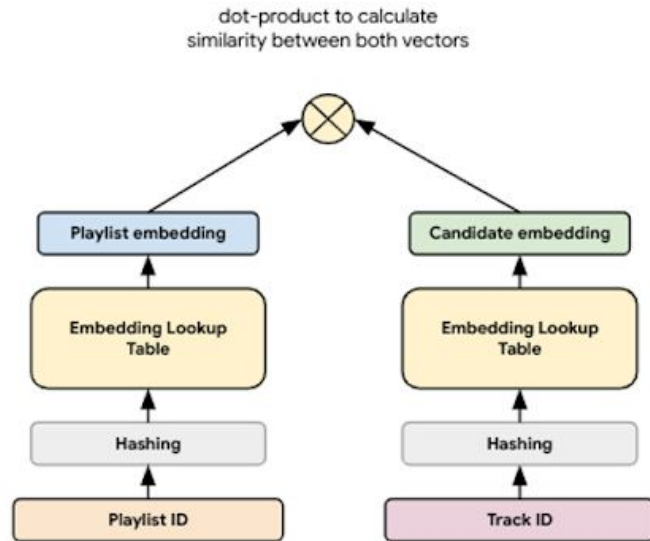
Deploy in production



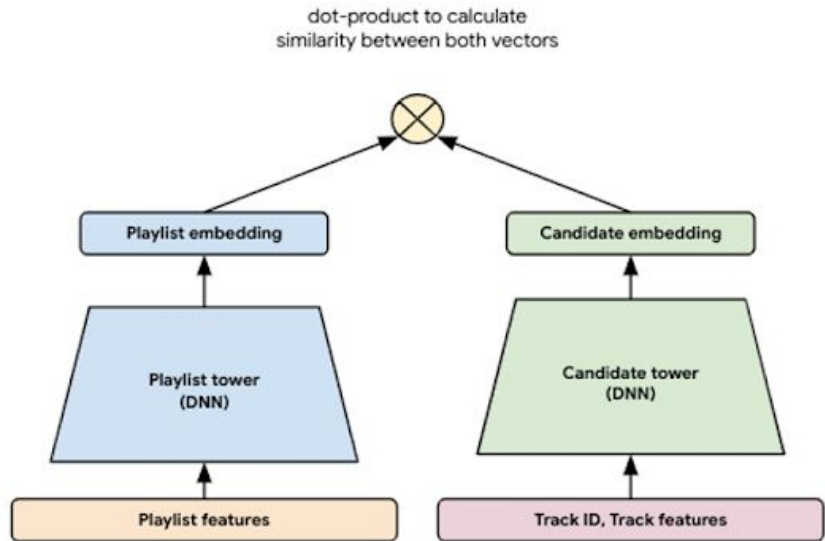
Two Tower Recommender



Two Tower Recommender



Factorization-based retrieval



Neural deep retrieval

TFX Pipeline Notebook

RecSys2025

> recommender_systems

> notebooks

> tfx_pipeline.ipynb



To be continued... in ~25 minutes



Up next: Running your own experiments

Part 3:

Experimentation

Some ideas you might like to try

Use alternative feature transformations

Add hidden layers, dropout, batch normalisation, etc. to each tower

Use bigger/smaller embedding dimensions

Transform and use additional product features



Thanks for attending!



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